YANNIK STAMM

PROGRAMMING, GAME DESIGN

GAMES PORTFOLIO

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ABOUT ME

Hi! I am Yannik
I am a 23 year old game developer from Hausen, Germany with a Bachelor in Games Engineering and I am currently pursuing my Master in Visual Game and Media Design in Copenhagen, Denmark.

I am a technical Game Designer with a strong focus on system-based game design, gameplay programming and game feel/feedback.

I primarily use Unity/C# but also have a strong understanding of C++.

This portfolio, showcases my game projects and some other game related projects.

My Skills

- Unity/C# (6 years)
- C++ (3 years)
- OpenGL (1 year)

- Reaper (5 years)
- Audacity (4 years)
- Blender (1 years)

- Systems & gameplay programming
- Game design & game feel
- Tools Programming

- Playtesting & user testing
- Rapid prototyping & iterative development

- Team-oriented & goal-driven
- Interdisciplinary work (3D-modeling, level building, music and sound design)
- Fluent in German & English
- Learning Danish

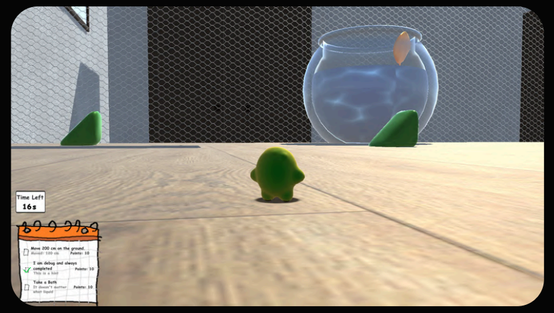
GAMES

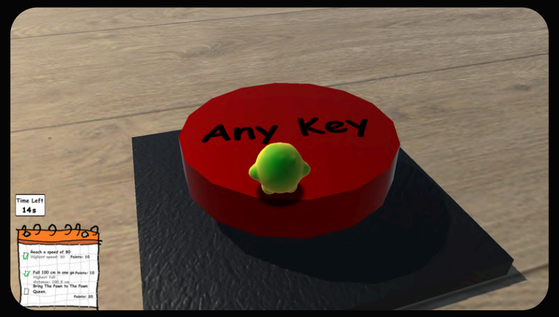

TONY BLOB'S
PRO BOUNCER

TONY BLOB'S PRO BOUNCER

Unity - Game Jam - "Change"









Bounce as Tony Blob through this sandbox level and succeed in versatile tasks.
After one minute, the tasks are randomly switched to new ones.

Time: 2 Weeks (2023) **Time size:** 2

: Gameplay

: Play it! (download)



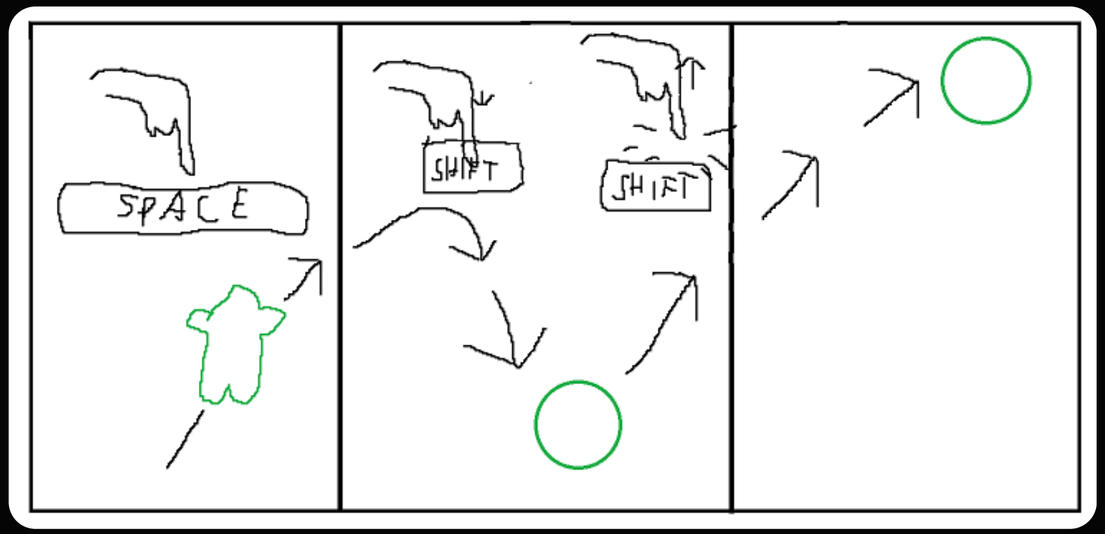
TONY BLOB'S PRO BOUNCER

The Concept
is to throw a lot of challenges at the player at the same time. We do this by giving the player a time limit and more tasks than he can fulfill.

The Level
is designed like a sandbox with stuff all over the place. Some of these objects help the player travel through the level quickly like the canons and the rubber slopes.



The Player Movement
was designed so that the player can move fast at one time and precise at another. I created a "bounce" mode, in which the player behaves like a rubber ball. This enables fast travels across the map and allows for increasing one's vertical speed by timing the bounce mode precisely.



BATECTIVE MITZI

"WILL SHE FIND THE PERPETRATOR AND GAIN THE OTHERS' TRUST?"

Yasha's scarf is missing, find out who stole it!

- Mitzi can talk, walk, climb walls and glide
- talk to others to find out their possible **motives,** but if they don't want to talk to a bat, *then*...
- ...fall onto animals and **bite** them to **take control** of their bodies
- get to the bottom of the **mystery** and save the day, no matter what it takes



BATECTIVE MITZI

Unity - Uni - Semester Project



You play as a bat that is somewhat new to the barn. The scarf of the barn's eldest went missing and you take it as your responsibility to find it again.

Time: 3 weeks (2025/26) **Team size:** 3

: Gameplay

: Play it! (browser)



BATECTIVE MITZI

The Main Character Controller
received most of my time. I wanted to create a player character that feels good to control and that interacts nicely with its environment.
The character controller can walk, climb, glide and take control of other animals.









BATECTIVE MITZI

Supervision

of the dialogue system was also one of my responsibilities. One of my team mates wanted to relearn computer science basics so we decided that he can create a custom dialogue parser. I supervised his work on that, gave him code reviews and discussed possible architectures on how to implement and integrate that system into the game.

```
Mitzi: Could you tell me about the others?
Yasha: Who would you like to know about?

+A: The horse
+B: The pigs
+C: The cat
#D: Nevermind

    >A
    Mitzi: What can you tell me about the horse?
    /* ... */
    >END

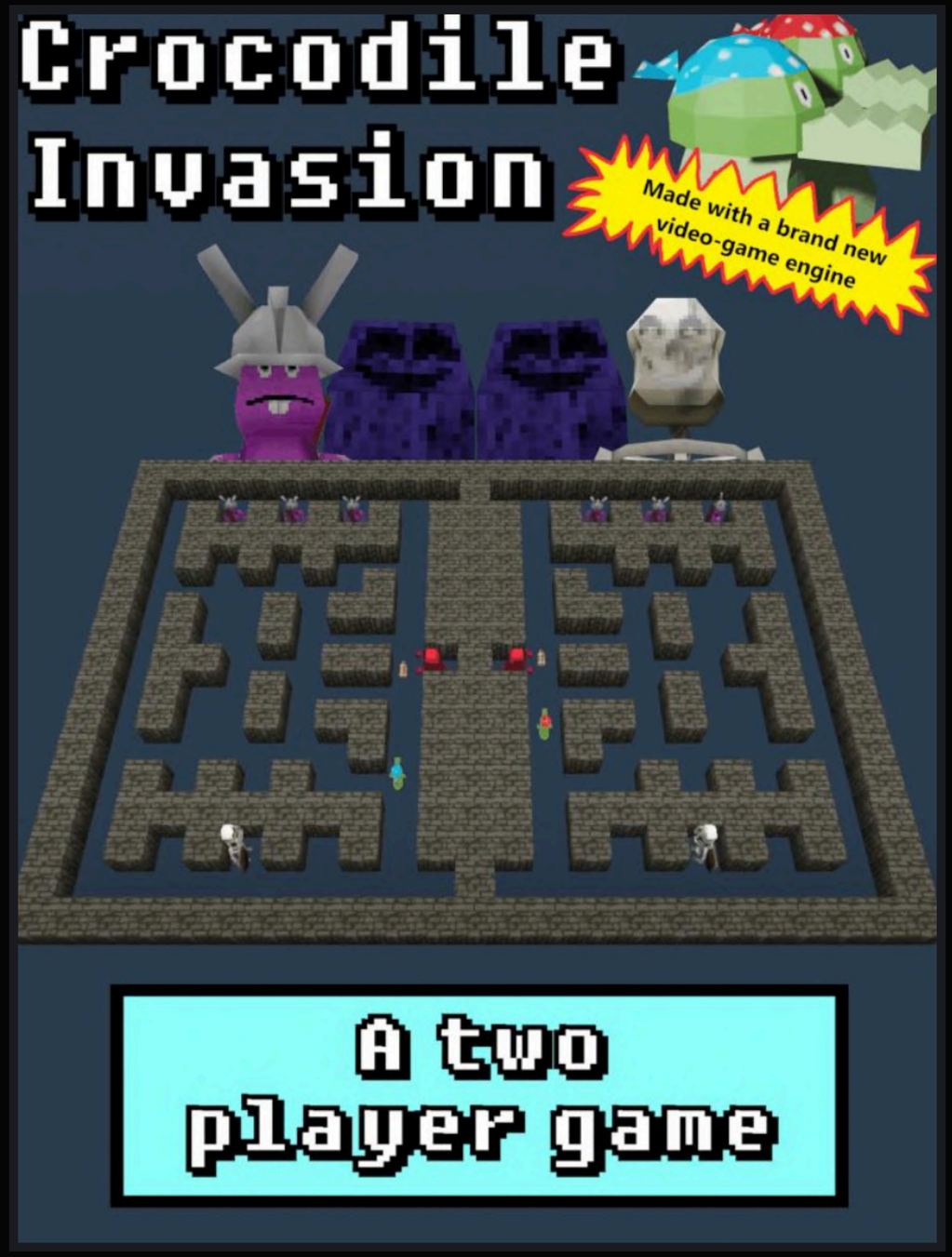
    >B
    Mitzi: Can you tell me about the pigs?
    /* ... */
    >END

    >C
    Mitzi: I hear a cat meowing. Do you know them?
    /* ... */
    >END
```

Talking Sounds

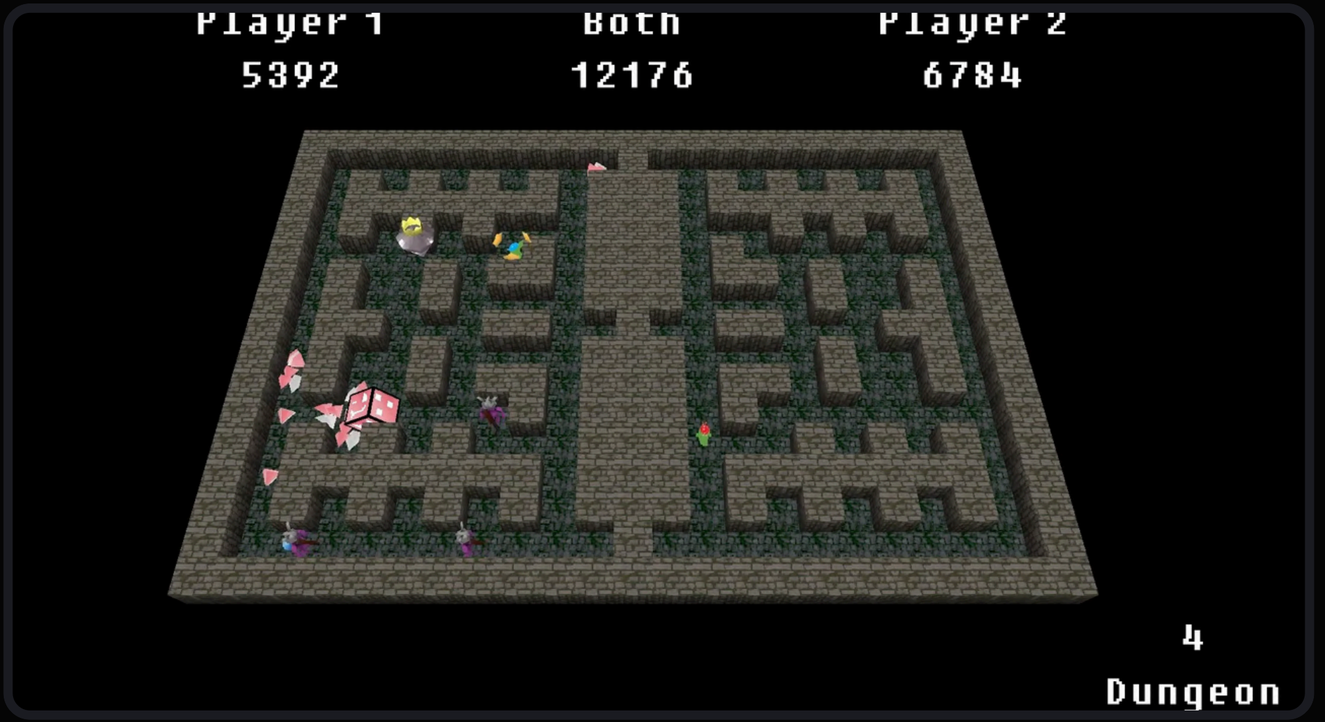
were also implemented by me. These are inspired by animal crossing's talk sounds and are just short samples that are continuously played in a random order during conversations. The system is simple, but it adds up to the game experience.




Crocodile Invasion
Made with a brand new video-game engine
A two player game

CROCODILE INVASION

Solo - University - C++/OpenGL



Multiplayer dungeon shooter that allows co-op and versus play styles.
Play through all 5 levels of the dungeon. Bring a friend or a foe to the party and either collaborate or compete over the highscore!

Time: Sep 2023 - Jul 2024 **Team size:** 1

CROCODILE INVASION

The Design Focus
was on making coop and versus playstyles enjoyable. Therefore, the game creates incentives for both playing styles.

Coop
- Defeating enemies is difficult alone
- Joined Score

Versus
- Only one can collect the points
- Separate Score

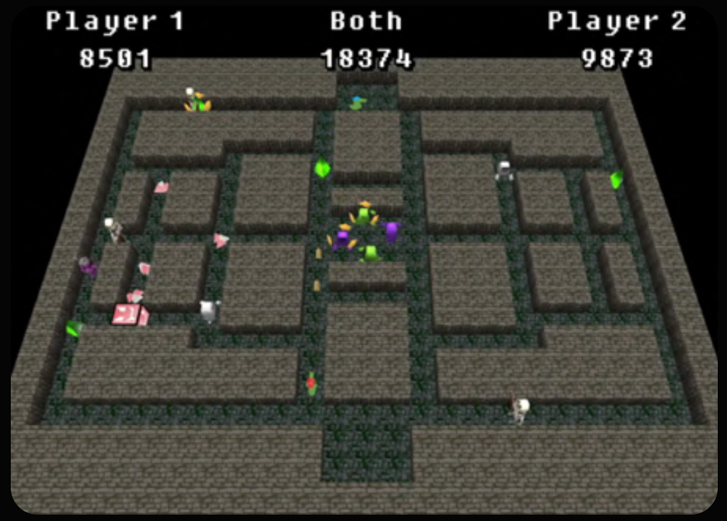
The Enemy Design
also reflects this. Most enemies are designed to be defeated together, but with more skill they can also be defeated alone.



: Gameplay

: Play it!

: Check the source code



CROCODILE INVASION

Technical Aspects

The base for Crocodile Invasionis a custom game engine, written by myself solely for this game with the following systems:

Data Systems
- ECS-System
- Inputhandler
- Scenes

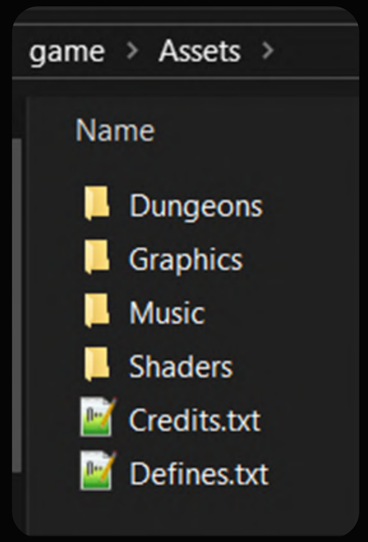
Graphical Systems
- glTF support
- Animations
- UI
- Text

Miscellaneous
- Collision detection
- Audio support

The ECS-System is the basis for most other systems, allowing a huge number of objects at the same time because of its efficient storage and cache usage.

The Game

itself has massive modding support. Close to all data can be replaced - from 3D-models, to dungeons and music.




Hungry Hungry Ant
Start Game

HUNGRY HUNGRY ANT

Unity - Game Jam - "End Where You Started"



Place arrows for the ant to guide it through the level. It must collect all leaves and bring them back to the colony.

Time: 48h (2024) **Team size:** 2

: Gameplay

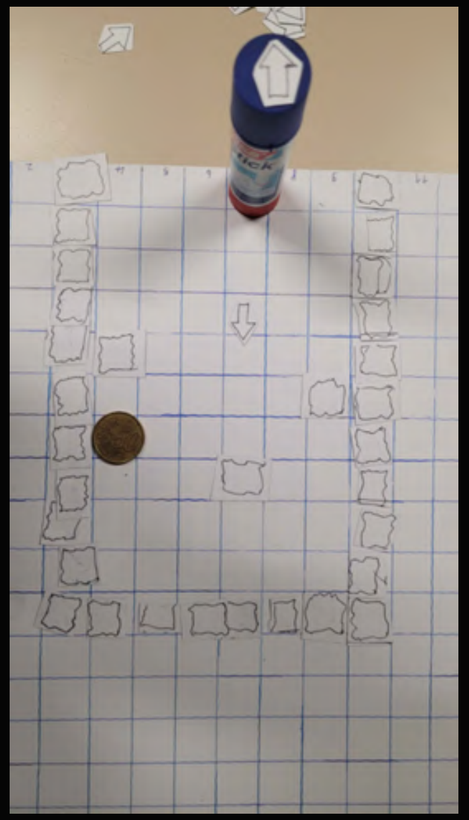
: Play it! (browser)



HUNGRY HUNGRY ANT

The Concept
is to have the player control an ant indirectly by placing arrows and stones. The ant will interact with these elements deterministically. The goal is to collect all leaves in the level and then have the ant return to the colony.

A Challenge
was the design of the levels. Figuring out a good level design and the number of items that are needed to solve a level was not so easy. To speed up the level creation process I created a paper prototype. Thus, we could just play around with one level idea, make a photo, and later add it into the game.



THE PHOEBUS CARTELL

Unity - Uni - Prototyping



You are a contract agent hired by the Phoebus cartel to destroy light bulbs so that the cartel can sell more of them.

Time: 2 days (2026) **Team size:** 3

: Play it! (browser)

The Concept

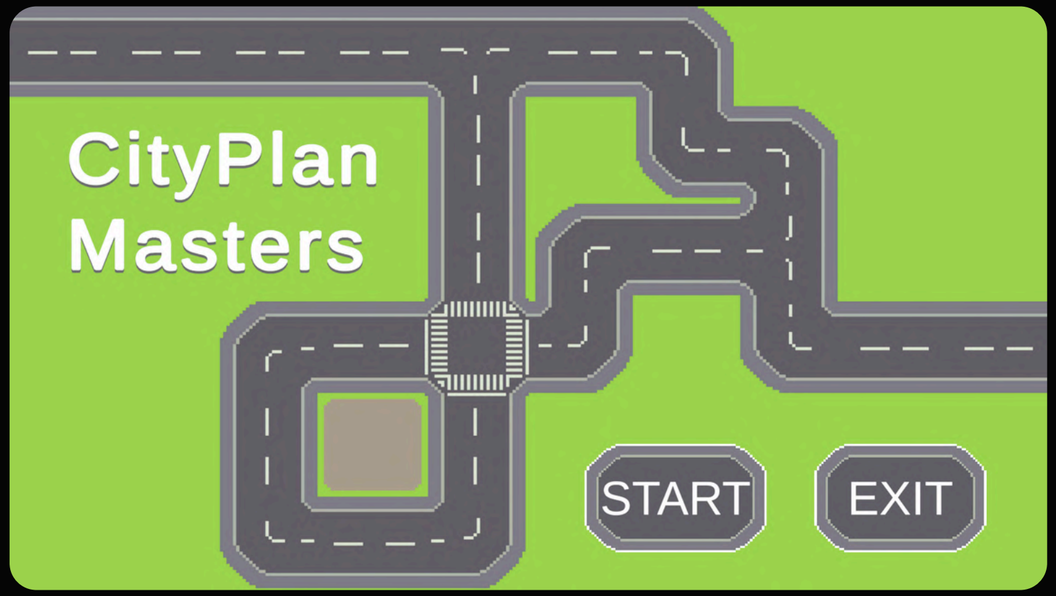
arose from the historical Phoebus cartel that supposedly downregulated the life span of light bulbs in order to sell more.

My Contributions

include the core idea of the gameplay, the implementation of the main gameplay loop and the controls. I also created some visuals like the window and the car.

CITYPLAN MASTERS

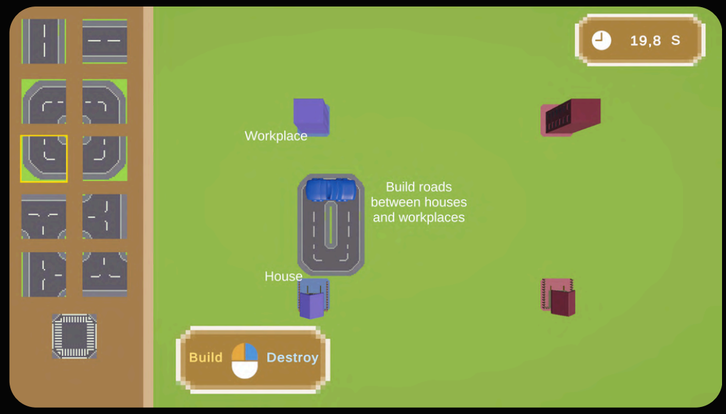
Unity - Uni - Prototyping



Solve simple traffic puzzles. Get all the people from their homes to their work places as fast as possible.

Time: 2 days (2026) **Team size:** 2

: Play it! (browser)



My Contributions

include the co-design of the gameplay loop and all work related to Unity and programmnig.


WUEHANS

wueHans

Eine Konsole aus Würzburg



Erlebe die Power von Open-Source

- WQVGA @ 60 fps
- Immersiver Klang
- 2 Controller

⇒ 2-facher Spaß



Spiel Spiele!

- Cowshooter
- Hans Tennis
- Hans Tower Defence

Entwickle Spiele!

WUE HANS

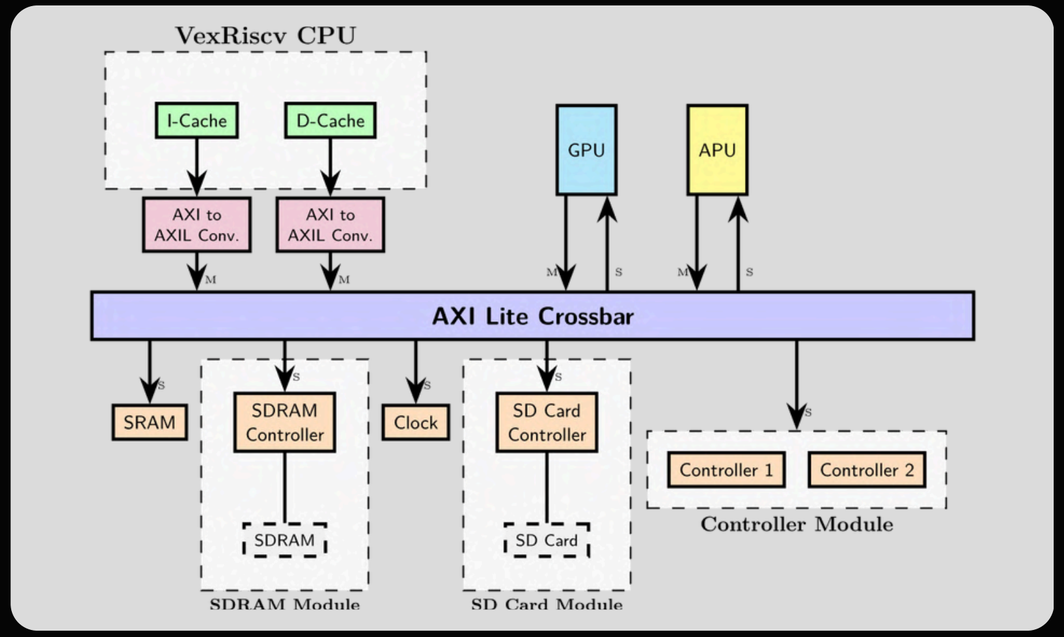
Hobby Project 2024/25

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WUE HANS

WUEHANS

Hobby - FPGA - Video Console



wueHans is an FPGA-based, custom video game console. It features a custom GPU and APU. You can write games for it using C++, and play them via an SD-Card on the console.

Time: Sep 2024 - Jul 2025 **Team size:** 3

: Presentation (German)

: Read the Wiki

My Role

was leading the project. I made sure that we stay in time while having all features ready that are needed.
I also developed core features like the GPU, most of the APU took responsibilities for key decisions in the architecture.

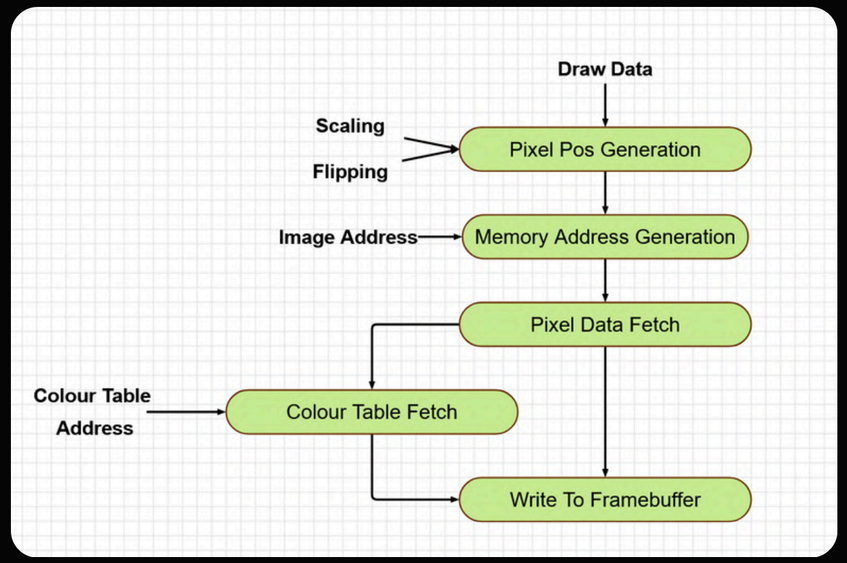
WUEHANS

The GPU
is a pipelined DMA with sprite related features in a nutshell. Those features are scaling, flipping and the optional use of colour tables.

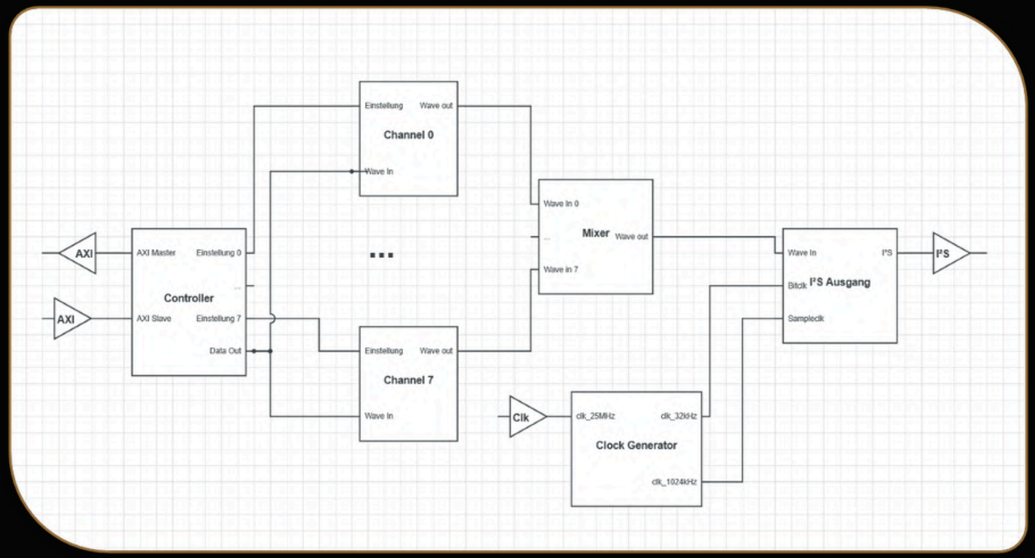
The CPU
is based on the VexRiscV, a Risc-V CPU generator. We adjusted it for max performance and wrote an AXI-Lite wrapper.

The APU
streams raw audio data from memory to the audio output. It has 8 independent channels that can be paired to form stereo channels.

GPU Pipeline Stages

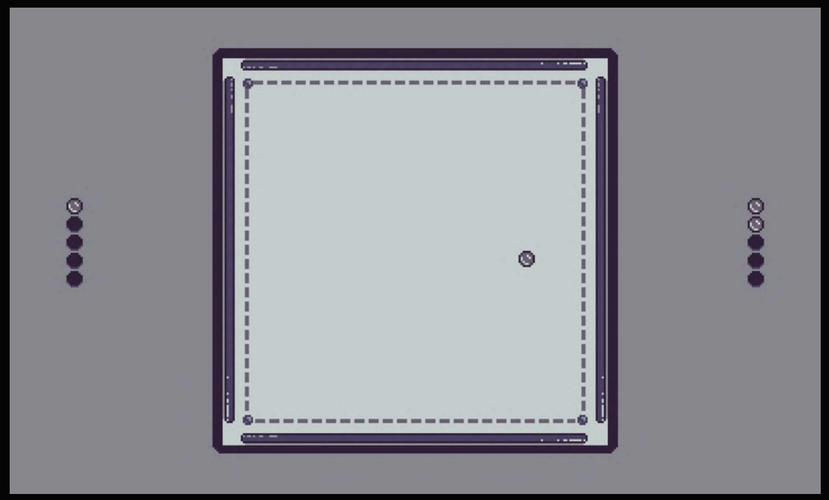


APU Architecture

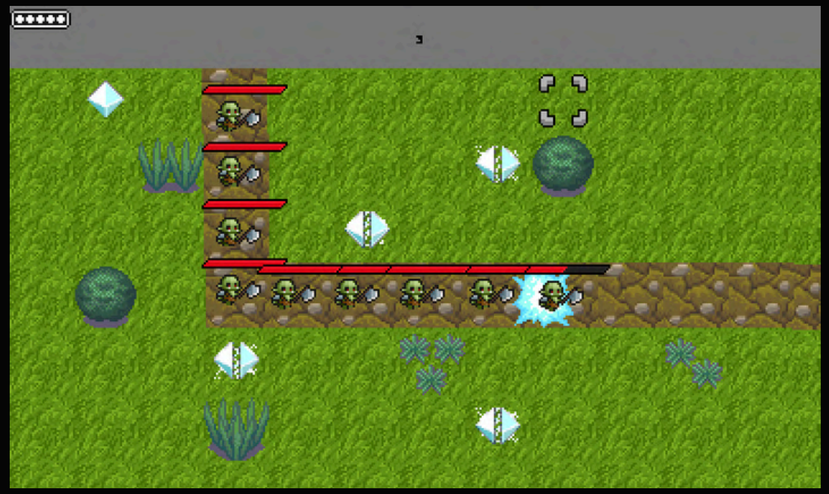


WUEHANS

Hans Tennis



Hans Tower Defence



Four Games

have been made for the console. Mostly during a small game jam that we hosted. We provided the participants with a software template, that includes a small game engine and an additional PC backend so that the participants can jump straight into making their game.

Hans The Gatherer



Cow Shooter



MUSIC

COMPOSITIONS AND ARRANGEMENTS

Reaper - Hobby - Video Gmes

Compositions

I also occasionally write music. Mostly in the style of video games and mostly for video games.

: Crocodile Invasion OST

: Tony Blob's Pro Bouncer OST

: My Soundcloud Profile

Arrangements

I also write arrangements for video game music that I like a lot. Usually it is either an orchrestal or a piano arrangement.

: Bowsers Inside Story - In The Final (Orchestral)

: Skylanders - Rumbletown (Piano)

: Kirby's Dreamland Medley (Strings + Piano)

THANK YOU!

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